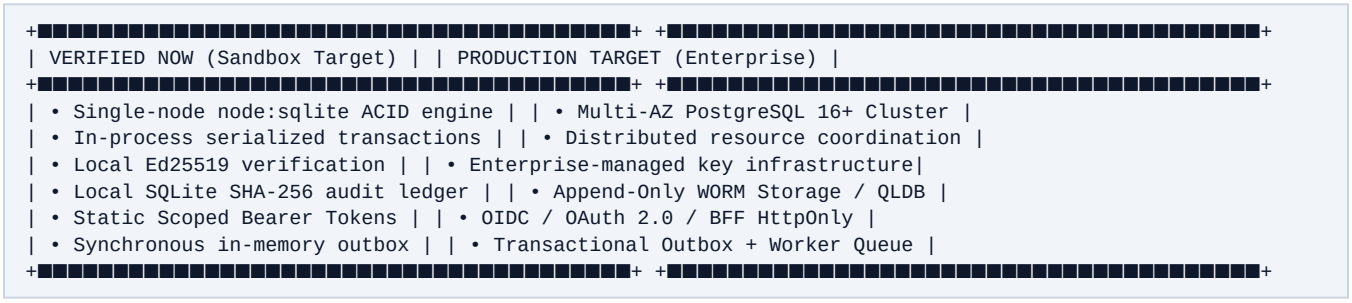
ACG Production Gap Analysis & Scaling Architecture

Sandbox Reference Implementation vs Enterprise Production Target

1. Scope & Architecture Separation



2. Production Scaling Statement

> ***Production scaling requires migration from the verified single-node SQLite reference implementation to a distributed persistence architecture such as PostgreSQL, with appropriate transactional resource coordination and enterprise-managed key-management infrastructure.***

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> ***The exact distributed locking and key-management technologies are deployment decisions rather than requirements of the ACG authorization model.***

3. Deployment Considerations (Options)

1. **Persistence:** PostgreSQL with row-level locking (`SELECT ... FOR UPDATE`) or distributed transaction managers.

2. **Resource Coordination:** Distributed transactional coordination (e.g., PostgreSQL advisory locks, Redis distributed locks, or etcd leases) selected based on deployment infrastructure.

3. **Key Infrastructure:** Enterprise-managed key management (e.g., AWS KMS, HashiCorp Vault, or cloud HSMs) for production signing and verification lifecycle.

4. **Session & Auth:** Enterprise OIDC / OAuth 2.0 / BFF HttpOnly cookie architecture for operator dashboard access.

5. **Durable Messaging:** Transactional outbox pattern for asynchronous fulfillment dispatch under network partitioning.